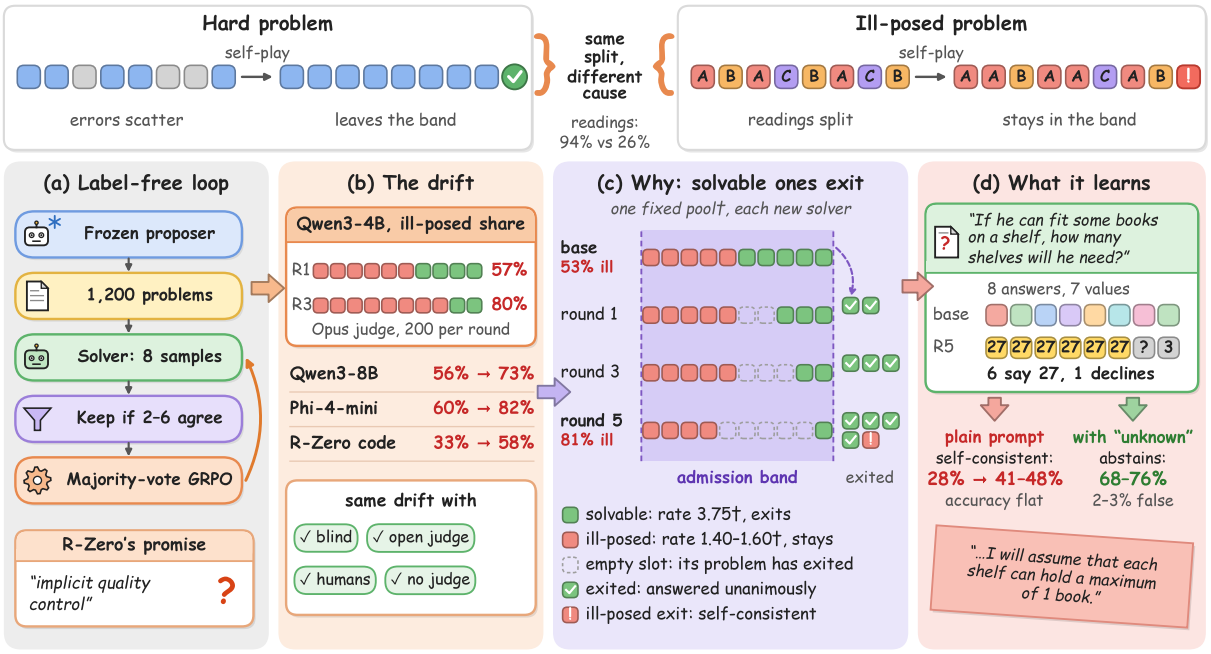

Hard problem
self-play
errors scatter
leaves the band
same split, different cause
readings: 94% vs 26%
Ill-posed problem
self-play
A B A C B A C B
A A B A A C A B !
readings split
stays in the band
(a) Label-free loop
Frozen proposer
1,200 problems
Solver: 8 samples
Keep if 2–6 agree
Majority-vote GRPO
R-Zero's promise
"implicit quality control"
?
(b) The drift
Qwen3-4B, ill-posed share
R1 57%
R3 80%
Opus judge, 200 per round
Qwen3-8B 56% → 73%
Phi-4-mini 60% → 82%
R-Zero code 33% → 58%
same drift with
✓ blind
✓ open judge
✓ humans
✓ no judge
(c) Why: solvable ones exit
one fixed poolt, each new solver
base
53% ill
round 1
round 3
round 5
81% ill
admission band
exited
solvable: rate 3.75†, exits
ill-posed: rate 1.40–1.60†, stays
empty slot: its problem has exited
exited: answered unanimously
ill-posed exit: self-consistent
(d) What it learns
"If he can fit some books on a shelf, how many shelves will he need?"
8 answers, 7 values
base
R5
27 27 27 27 27 27 ? 3
6 say 27, 1 declines
plain prompt
self-consistent:
28% → 41–48%
accuracy flat
with "unknown"
abstains:
68–76%
2–3% false
"...I will assume that each shelf can hold a maximum of 1 book."